

(d) One solution of the equation $2z^3 - 5z^2 + cz - 5 = 0$ is $z = 1 - 2i$.

If c is real, find the value of c and the other two solutions of the equation.

Another solution is the complex conjugate of the given solution because the coefficients are real, so we can multiply these factors together:

$$\begin{aligned}(z - 1 + 2i)(z - 1 - 2i) \\&= z^2 - 2z + 1 + 4 \\&= z^2 - 2z + 5\end{aligned}$$

$$\begin{aligned}(2z - 1)(z^2 - 2z + 5) \\&= 2z^3 - 4z^2 + 10z - z^2 + 2z - 5 \\&= 2z^3 - 5z^2 + 12z - 5\end{aligned}$$

$$c = 12$$

$$z_2 = 1 + 2i$$

$$z_3 = \frac{1}{2}$$